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A pilot study on social science graduate students' data core competency

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ABSTRACT

In the field of research, teaching graduate students how to manage data on a regular basis is becoming increasingly crucial. However, there is a lack of adequate evidence in the literature to determine the necessity of Research Data Management (RDM) training from the perspectives of social science student researchers. To address this gap, a pilot study was conducted to develop a data core competency measure to gauge the perceived importance of data competency among social science graduate students and assess their data core competency. Forty graduate students in a college of education at an US research university filled out a researcher-designed data core competency survey. This pilot study found that graduate students in the social sciences do not generally appreciate RDM. Consequently, the results underscore the need to foster an environment in graduate social science programs that value RDM best practices. The study suggested that future studies build on the preliminary findings by developing standardized data core competency measures.

Introduction

Rapid changes in technology over the last ten years have completely changed how researchers produce, collect, and analyze research data. As a result, today's scholars and researchers are expected to possess up-to-date knowledge and skills in Research Data Management (RDM) to succeed in the field. Broadly speaking, RDM refers to the effective management of research data throughout the entire research cycle, from data entry to distribution and to archiving results (Wilms et al., 2020).

RDM has become a more significant factor in studies over the last decade (Belter, 2014; Wilson & Jeffreys, 2013; Xu et al., 2022b). Many research stakeholders now support, encourage, and even mandate research data sharing. Additionally, appropriate data management is essential to enable data sharing. Well-documented and preserved data is easily accessed, interpreted, and used by other interested researchers. When informational data and materials can be incorporated into research publications, it makes research more efficient (Kuula & Borg, 2008). There has been a longstanding need to provide ongoing data management instruction for graduate students (Johnston & Jeffryes, 2014; Raszewski et al., 2021; Williams, 2013).

Gradually, in recent years, a larger number of academic libraries have established services to support the development of graduate students' RDM skills (Tenopir et al., 2015). However, RDM instruction for graduate students is still in an early phase (Xu, 2022; Xu et al., 2022b). For instance, only a few studies reported specifically tailored RDM instruction for graduate students in social science disciplines, yet the graduate students expressed a willingness to receive more instruction and RDM resources specifically tailored for their unique discipline (Johnston & Jeffryes, 2014; Whitmire, 2015).

Understanding students' experiences with and perceptions of research data management is of critical importance to providing them with training. Although several studies have attempted to identify the required skill sets for researchers across various disciplines and fields (Gurcan, 2019; Padilla, 2016), the literature lacks sufficient evidence on RDM instruction needs from the perspectives of student researchers or their level of preparedness in data core competency, particularly in the social sciences. Therefore, the present study aims to address this research gap by evaluating social science graduate students' perceptions of the significance and preparedness of their data core competency.

Literature review

RDM challenges

The literature has identified several challenges and barriers surrounding RDM and data sharing (Borgman, 2010; Koltay, 2017;

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